

XAI-EdgeSFL: Explainable Edge Intelligence With Adaptive Intrusion-Resilient Split Federated Learning for Consumer Healthcare Ecosystems

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Abstract—The fast growth of consumer electronics, particularly Internet of Things (IoT) devices and wearables, has significantly changed healthcare systems by enabling real-time monitoring and customized medical treatment. Edge computing is changing healthcare by enabling the secure and distributed processing of data on consumer devices. However, this integration presents substantial security issues, raising significant concerns regarding AI-driven decision-making, model understandability, and privacy protection. This paper introduces XAI-EdgeSFL, a novel framework that integrates Explainable Artificial Intelligence (XAI) with an Adaptive Intrusion-Resilient Split Federated Learning (SFL) architecture to provide reliable and transparent edge health monitoring. The SFL loop computes important features of sensitive health data using a lightweight Temporal Convolutional Network (TCN) on edge devices and a more complex classifier on a global server. This method enhances privacy by retaining raw data on the device. In addition, we present an adaptive aggregation approach that dynamically weights client model updates to evaluate intrinsic resilience against emerging cyberthreats. SHapley Additive exPlanations (SHAP), Local Interpretable Model-agnostic Explanations (LIME), and Integrated Gradient (IG) are a multi-modal XAI suite that offers feature-level explanations for clinical trust. The proposed method performs well across all datasets, achieving 86.15% accuracy and 86.01% F1-score on CICIoMT2024, 93.73% accuracy and 93.65% F1-score on N-BaIoT, and 99.46% accuracy and 99.47% F1-score on CICSIDS2017. The results show that XAI-EdgeSFL improves security, privacy, and transparency for AI-driven consumer healthcare systems.

Index Terms—Explainable artificial intelligence, split federated learning, edge intelligence, adaptive intrusion detection, consumer healthcare systems, security & privacy.

I. INTRODUCTION

THE rapid digital transformation in healthcare, driven by IoT, has allowed for continuous patient monitoring and highly personalized care. The development of interconnected ecosystems that generate a continuous stream of healthcare data has been facilitated by the extensive adoption of consumer electronics, including smartwatches, fitness monitors, and medical IoT devices [1]. Decentralized health surveillance has the potential to substantially improve clinical outcomes by using real-time data analysis and proactive intervention strategies.

The concept of edge computing has become a leading paradigm for facilitating efficient, privacy-preserving, distributed data processing near consumer devices. It reduces latency and saves bandwidth by allowing local analysis, which helps to solve privacy concerns that arise when sensitive medical information is frequently transmitted to centralized cloud systems. However, the decentralized and broad presence of healthcare ecosystems makes them vulnerable to cyberattacks [2]. An effective intrusion may compromise or modify important health alarms, which may directly harm patient safety. It, therefore, becomes imperative to prioritize the security of such systems. The centralized structure of traditional Intrusion Detection Systems (IDS) may have difficulties adapting to this environment. These systems face two main challenges [3]. First, they need to handle distributed data without violating the key privacy principles of edge computing. Second, the low explainability of modern AI-based models often leads to closed system decision-making, which provides little understanding of the rationale behind specific activities, which are labeled malicious.

Federated Learning (FL), which supports the joint training of models across devices and protects raw data, is a way of addressing data privacy issues. Nevertheless, traditional FL methods may be computationally heavy to run on edge devices with limited resources, which makes them susceptible to model poisoning attacks by malicious agents [4]. SFL presents a distinct benefit of dividing the model into client and server

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parts. This approach reduces the computing risks on edge devices and avoids the transmission of harmful activities to final layers [5] [6] [7]. In spite of these advantages, even globally aggregated models that are trained on non-IID (non-independent, identically distributed) data do not generally have interpretable outputs. The obscurity of such AI-based systems in the clinical environment impedes the capacity of healthcare professionals to trust and respond to the automated security warnings. XAI aims to increase trust by enhancing model interpretability and accountability [8]. XAI developments have been focused on centralized learning frameworks, leaving distributed systems such as SFL largely unexplored. XAI-EdgeSFL handles these issues directly by merging adaptive, intrusion-resistant SFL with multi-modal XAI. Data is handled locally using lightweight TCNs for edge efficiency, with adaptive aggregation and trust scoring to resist model poisoning. For clinical transparency, feature-level explanations are provided using SHAP, LIME, and integrated gradients.

The proposed XAI-EdgeSFL framework enhances security, privacy, and transparency in healthcare edge systems. It distributes tasks between local TCN models and a central server for efficiency. The design improves resistance to model poisoning and uses XAI tools like SHAP for interpretability, increasing clinical trust. The following are the main research contributions.

- 1) We present XAI-EdgeSFL, an innovative architecture for consumer healthcare devices with limited resources, which integrates XAI with adaptive intrusion-resilient SFL. The security and transparency challenges in distributed healthcare systems are addressed by this integrated approach.
- 2) We develop a lightweight TCN for deployment on resource-constrained edge devices. This design enables effective temporal feature extraction from sequential network traffic while retaining raw data on the client device, thereby protecting user privacy and reducing the computational burden on consumer devices.
- 3) We present an adaptive aggregation method for SFL that adjusts the weights of client contributions according to performance measures in real-time. This method effectively mitigates model cyberattacks by limiting the impact of suspicious clients while improving convergence through prioritized learning from trustworthy clients.
- 4) We develop an extensive multi-modal XAI suite that includes SHAP, LIME, and Integrated Gradients. This method addresses interpretability issues in distributed learning systems and enhances clinical trust in AI-driven security decisions by providing feature-level explanations for model predictions.

The remaining part of the paper is organized as follows: Section II explains the related work, and Section III describes the proposed method. Section IV presents the experimental results, and Section V concludes the work.

II. RELATED WORK

Traditional machine learning-based methods of threat detection in healthcare devices are rapidly being substituted by

edge-based FL approaches. This progression solves major shortcomings of centralized approaches, notably with regard to data privacy and the computational restrictions of consumer medical devices [2], [6], [9]. The following subsections critically evaluate several security measures, focusing on the shift from cloud-based to distributed intelligence employing edge computing, FL, and SFL.

A. Edge Computing

Lin et al. [10] examine the allocation of resources in edge computing and the architecture of IDS. The proposed system effectively optimizes the allocation and sharing of heterogeneous resources. A single-layer dominant and max-min fair (SDMMF) allocation technique ensures hierarchical fairness within a broad edge-based IDS. Allocation of multiple resources across layers is handled using a method called Multilayer Dominant and Max-Min Fair (MDMMF). Cui et al. [11] improve edge computing intrusion detection using gradient boosting decision trees by addressing data imbalance, high dimensionality, and inefficient parameter tuning. The offered approach uses Margin SMOTE to introduce balanced data generation and a recursive feature removal hierarchy to select features. Also, a grid search method is optimized to a customized one to optimize the efficiency of the parameters. Edge computing technology brings significant time improvements to a real-time environment and occupies less bandwidth. Singh et al. [12] suggest the creation of a smart healthcare system based on an edge computing paradigm. The solution proposes an intermediate edge layer that minimizes the network latency, and patient data privacy is guaranteed by using privacy-preserving searchable encryption and an access control system to validate illegal access to data. Eskandari et al. [13] introduce a smart edge-based intrusion detection system that aims at protecting IoT edge devices directly attached to it. The system applies to low-cost IoT gateways that assume edge computing to detect vulnerabilities that are close to the data sources. It identifies a large number of threats, such as port scanning, HTTP and SSH brute force, and SYN flood, with high accuracy and a low rate of false positives. Yao et al. [14] proposed an edge-based anomaly detection system to detect malicious attacks. A wrapper-based feature selection method eliminates irrelevant and duplicate features, thus improving detection capabilities, and four types of attackers are distinguished based on their capabilities. Ensemble learning is used to increase levels of detection, and a lightweight algorithm is also used to minimize latency by running it at the same time on edge nodes.

B. Federated Learning

Several research studies [15], [16] used FL approaches in the healthcare domain. Mothukuri et al. [4] implemented an anomaly detection methodology that is based on FL and proactively detects IoT-based intrusions by analyzing data from each device. This system utilizes FL cycles with GRUs, preserves local data, and exclusively transmits learned weights to the global server. The results show that decentralized solutions perform better in terms of accuracy and privacy

preservation. Alamleh et al. [17] standardize and benchmark machine learning-based IDS in FL for healthcare applications using a multicriteria decision-making framework. A decision matrix is generated by utilizing a dataset of 125,973 records and 41 attributes, and the fuzzy Delphi technique is employed to standardize evaluation criteria. Borda voting is used to assign weights and choose the best solution. Hallaji et al. [18] investigate the various forms of dangers and adversaries that decentralized FL encounters, as well as its security mechanisms, dependability, and authenticity. FL is recognized for its privacy-preserving features, but sharing model updates can compromise model performance and data privacy. Decentralized FL aims to minimize server-related risks by leveraging blockchain technology; however, this raises additional privacy concerns. Santos et al. [19] introduce an FL model for a reliable network-based IDS with high-confidence model updates over time. Unsupervised classification reliability evaluation in a federated architecture eliminates probable misclassifications while lowering model update costs. Using the MAWIFlow dataset, experiments reduced false-positive and false-negative rates by up to 12% and 9.6%, respectively. They also improved false-positive reduction and reduced update costs. Pedro et al. [20] evaluate the efficiency of differential privacy approaches in training an FL-based IDS for industrial IoT. The ToN_IoT dataset's non-IID data is used to compare privacy levels and aggregation methods, including FedAvg and Fed+. The results show that Fed+ maintains similar precision even when noise is introduced during federated training.

C. Split Federated Learning

Thapa et al. [21] introduce SFL, a hybrid of FL and Split Learning (SL) that combines their advantages while mitigating their limitations. Relay-based client coordination slows training in SL, but it provides better model privacy and suits resource-constrained scenarios. However, SFL considerably reduces computing time per global epoch for multiple clients while retaining accuracy and communication efficiency. Data privacy and model robustness are enhanced by differential privacy and PixelDP. Experiments show that SFL scales adequately with client count and works well in privacy-preserving scenarios [6]. Otoum et al. [22] suggest that the SL approach is effective in achieving a higher detection rate, accuracy, and classification performance for an IDS. The prospective security concerns have been addressed by employing the SL-based IDS for advanced transportation systems. The proposed model has been analyzed and compared to other models, including FL and transfer learning solutions. Compared to FL and transfer learning, the suggested SL achieves a 2–5% improvement in accuracy and a 5% improvement in detection rates, respectively. Turina et al. [23] propose SFL, a hybrid of federated and SL that incorporates their privacy and efficiency advantages. SFL has a superior accuracy-privacy tradeoff than parallel SL, allows for concurrent processing, and minimizes client processing requirements. Additionally, it maintains high accuracy on imbalanced data. Pasquini et al. [24] examine the security implications of SL, a collaborative machine learning approach that aims for high performance and low resource

TABLE I
CLASS DISTRIBUTION ACROSS DATASETS

Dataset	Class	Number of Samples (%)
CICIoMT2024	Benign	311 (3.11%)
	DDoS	4,881 (48.81%)
	DoS	3,965 (39.65%)
	MQTT	587 (5.87%)
	Recon	239 (2.39%)
N-BaIoT	Spoofing	17 (0.17%)
	Ack	57,997 (13.9%)
	Scan	97,096 (23.2%)
	Syn	61,851 (14.8%)
	UDP	153,414 (36.7%)
CICIDS2017	UDP Plain	47,409 (11.3%)
	Benign	5,202 (41.7%)
	Brute Force	5,716 (45.9%)
	SQL Injection	36 (0.3%)
	XSS	1,507 (12.1%)

consumption. This framework contains numerous vulnerabilities that enable attackers to reconstruct clients' private training data, according to the research. A compromised or hostile server can modify distributed learning, exposing client data in a vulnerable model state. Several datasets are used to test different attack variations under realistic threat scenarios. The CPU capacity of edge devices, such as IoT devices, is insufficient to manage large machine learning models with millions of parameters. Client-server confidentiality during training and testing is also critical. SL addressed these concerns by dividing the model between clients and the server, allowing each to train its own section without communicating data or models [25]. SFL integrates the privacy and model-splitting benefits of SL with the speed of FL.

This paper introduces XAI-EdgeSFL, a secure and transparent edge healthcare architecture that integrates adaptive intrusion-resilient SFL and XAI. A lightweight TCN on edge devices extracts local features while a global server performs complex classification tasks, saving raw data on the devices for data privacy. An adaptive aggregation technique enhances resilience to cyberattacks, while a multi-modal XAI suite presents interpretability for clinical trust. The experimental results show that XAI-EdgeSFL enhances privacy, security, and transparency in AI-powered healthcare systems.

III. PROPOSED METHOD: XAI-EDGESFL FRAMEWORK

Figure 1 illustrates that XAI-EdgeSFL is a secure edge health monitoring solution that integrates adaptive, intrusion-resilient SFL with XAI. TCN extracts features on clients, thereby preserving the confidentiality, while the classifier is executed on the global server. Adaptive aggregation improves resilience, whereas SHAP, LIME, and integrated gradients provide feature-level explanations. The method architecture is discussed in the following subsections.

A. Dataset Preparation

The proposed approach was evaluated using three realistic datasets. CICIoMT2024¹ [26] targets healthcare IoT devices,

¹<https://www.unb.ca/cic/datasets/iomt-dataset-2024.html>

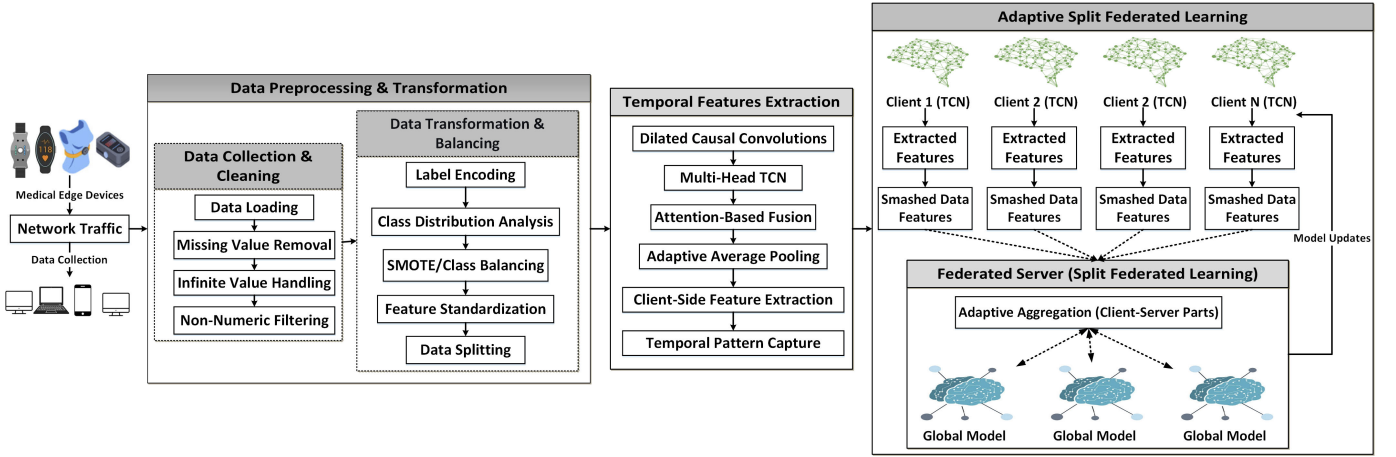


Fig. 1. XAI-EdgeSFL: Proposed system illustrating the integration of XAI with Adaptive Intrusion-Resilient SFL for consumer healthcare ecosystems.

simulating attacks on 40 devices (25 real, 15 virtual) communicating via Wi-Fi, MQTT, and Bluetooth. It includes five attack types: DDoS, DoS, reconnaissance, MQTT-based attacks, and spoofing. Traffic was captured through a network tap, generating a dataset by replaying real-time packets between the switch and IoMT devices. N-BaIoT² [27] was created in a laboratory by infecting nine commercial IoT devices with Mirai botnet traffic, enabling evaluation of the method's ability to detect malicious communications from compromised devices. Finally, CICIDS2017³ [28] solves the constraints of synthetic IDS datasets by presenting legitimate benign traffic alongside recent attacks. The B-Profile system imitates user actions across many protocols. It generates labeled PCAPs and flow-based CSV files, which are then used for accurate attack evaluation. Table I shows the class distribution for the CICIoMT2024, N-BaIoT, and CICIDS2017 datasets. It illustrates the class imbalance in the datasets by showing the quantity and proportion of samples for each attack and benign class. For instance, CICIoMT2024 is primarily composed of DDoS and DoS attacks, N-BaIoT primarily consists of UDP traffic, and CICIDS2017 has a higher proportion of benign and brute-force samples.

B. Data Preprocessing and Feature Engineering

The data preprocessing pipeline starts by loading the dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^M$, where y_i indicates class labels and $x_i \in \mathbb{R}^d$ indicates feature vectors. To maintain data integrity, the system cleans all rows with missing or infinite values using $\mathcal{D}_{\text{clean}} \leftarrow \text{RemoveNaN}(\mathcal{D})$. The framework employs SMOTE (Synthetic Minority Over-sampling Technique) to mitigate the significant problem of imbalanced class proportions in intrusion detection datasets. This technique generates synthetic samples for minority classes to generate $\mathcal{D}_{\text{balanced}} \leftarrow \text{SMOTE}(\mathcal{D}_{\text{clean}})$ [29], [30]. This method generates new instances by interpolation using equation 1.

$$\hat{x} = x_i + \lambda \cdot (x_j - x_i) \quad (1)$$

where $k = \min(5, n_{\text{minority}} - 1)$ is the adaptive parameter setting, x_i is an instance of the minority class, x_j is one of its k -nearest neighbors, and $\lambda \sim U(0, 1)$. The application of SMOTE is appropriate due to its simplicity, computational efficiency, broad adoption within the research community, and its capability to address class imbalance in feature-based learning environments. The TCN architecture suggested processing fixed-length sequential segments separately and concentrates on local patterns rather than long-term components. SMOTE provides an effective and realistic approach to balancing data without compromising detection accuracy. Following class balancing, equation 2 is used to apply feature standardization.

$$\tilde{x}_i \leftarrow \frac{x_i - \mu}{\sigma} \quad (2)$$

where μ and σ represent the mean and standard deviation of each feature dimension, respectively. This transformation assures $\mathbb{E}[\tilde{x}] = 0$ and $\text{Var}[\tilde{x}] = 1$, allowing for enhanced model convergence during training. The data is converted into a sequential format using a sequence length L during the preprocessing procedure, which is critical for identifying network traffic behaviors during the temporal analysis. The preprocessing stage involves data segmentation using an 80-20 stratified split, followed by client distribution using equations 3 and 4.

$$\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}} \leftarrow \text{TrainTestSplit}(\mathcal{D}_{\text{balanced}}) \quad (3)$$

$$\{\mathcal{D}_k\}_{k=1}^N \leftarrow \text{IIDPartition}(\mathcal{D}_{\text{train}}, N) \quad (4)$$

This distribution strategy maintains statistical properties across clients, supporting robust FL performance. It is critical to clarify the experimental design regarding data distribution. The TCN architecture and adaptive aggregation, which are essential processes in the XAI-EdgeSFL system, are investigated in the context of an IID partition. This is a controlled baseline that allows for easy separation of model performance and the confounding variable of data heterogeneity. Sections I and V explore the significance of our adaptive aggregation technique in addressing non-IID data challenges in real-world federated healthcare implementation.

²<https://ieee-dataport.org/documents/n-baiot>

³<https://www.unb.ca/cic/datasets/ids-2017.html>

Algorithm 1 Multi-Head TCN Client Update

Input: Client k , global parameters Θ_c, Θ_s , sequence length L , heads H

Output: Updated parameters $\Theta_c^{(k)}, \Theta_s^{(k)}$, attention weights A

Client-side Multi-Head TCN Forward Pass:

Initialize: $\Theta_c^{(k)} \leftarrow \Theta_c, \Theta_s^{(k)} \leftarrow \Theta_s$

Prepare sequences: $\mathcal{B} \leftarrow \text{SequenceBatch}(\mathcal{D}_k, L)$

for each batch $(X_b, y_b) \in \mathcal{B}$ **do**

Multi-Head TCN Processing:

Reshape: $X'_b \leftarrow \text{reshape}(X_b)$

Project: $Z \leftarrow W_{\text{proj}} X'_b$

Multi-Head Temporal Processing:

Initialize: $\mathcal{H} \leftarrow \{\}$

for $h = 1$ to H **do**

$\text{dilation}_h \leftarrow 2^{h-1}$

$T_h \leftarrow \text{TemporalBlock}(Z, \text{dilation}_h)$

$h_h \leftarrow \text{GlobalAvgPool}(T_h)$

$\mathcal{H} \leftarrow \mathcal{H} \cup \{h_h\}$

end

Attention Mechanism:

$H_{\text{concat}} \leftarrow \text{concat}(\mathcal{H})$

$\alpha \leftarrow \text{softmax}(W_2 \tanh(W_1 H_{\text{concat}}))$

$h_{\text{final}} \leftarrow \sum_{h=1}^H \alpha_h \cdot h_h$

Server-side Classification:

$\hat{y}_b \leftarrow f_s(h_{\text{final}}; \Theta_s^{(k)})$

Optimization:

$\mathcal{L} \leftarrow \text{CrossEntropy}(\hat{y}_b, y_b) + \lambda \|\Theta\|_2^2$

Update: $\Theta_c^{(k)} \leftarrow \Theta_c^{(k)} - \eta \nabla_{\Theta_c^{(k)}} \mathcal{L}$

Update: $\Theta_s^{(k)} \leftarrow \Theta_s^{(k)} - \eta \nabla_{\Theta_s^{(k)}} \mathcal{L}$

end

return $\Theta_c^{(k)}, \Theta_s^{(k)}, A, \{\phi_{IG}, \phi_{\text{attention}}\}$

TABLE II

TCN ARCHITECTURE HYPERPARAMETERS

Parameter	Value	Description
Sequence Length (L)	10	Input temporal window size
TCN Heads (H)	4	Parallel dilated convolution heads
Kernel Size (k)	3	Convolutional filter dimension
Hidden Channels	64	Feature dimension in convolution
Dropout Rate (ρ)	0.3	Regularization probability
Dilation Rates	[1, 2, 4, 8]	Exponential dilation progression
TemporalBlocks/Head	2	Depth of each TCN head
Batch Size	32	Training samples per iteration
Learning Rate (η)	0.001	Adam optimizer step size

where Conv1D_d is dilated convolution, BN is batch normalization, and Proj is a 1×1 convolution (linear projection) that matches dimensions for the residual connection when input and output channel counts differ. The architecture follows standard TCN designs with optimizations for edge computing constraints. The network initiates with input reshaping, as indicated by the expression $X'_b \leftarrow \text{reshape}(X_b)$. Subsequently, an input projection layer is implemented using equation 6.

$$Z \leftarrow W_{\text{proj}} X'_b \quad (6)$$

The adaptable projection matrix, which converts the input features into a representation suitable for temporal processing, is denoted by W_{proj} . The main innovation is the multi-head TCN design that uses H parallel temporal blocks with 2^{h-1} dilation rates that increase exponentially from $h = 1$ to H [32]. Equation 7 is employed by each head to analyze the estimated input Z through temporal blocks.

$$T_h \leftarrow \text{TemporalBlock}(Z, \text{dilation}_h) \quad (7)$$

Subsequently, the global average pooling $h_h \leftarrow \text{GlobalAvgPool}(T_h)$ is carried out to obtain concise temporal representations. The multi-head outputs are accumulated in $\mathcal{H} \leftarrow \mathcal{H} \cup \{h_h\}$ and subsequently integrated through an attention procedure that employs equation 8.

$$\alpha \leftarrow \text{softmax}(W_2 \tanh(W_1 H_{\text{concat}})) \quad (8)$$

Here, $H_{\text{concat}} \leftarrow \text{concat}(\mathcal{H})$ and α indicate dynamic attention weights for distinct temporal scales. Equation 9 provides the final client representation.

$$h_{\text{final}} \leftarrow \sum_{h=1}^H \alpha_h \cdot h_h \quad (9)$$

This design efficiently extracts features while ensuring privacy by processing raw data locally and sending just the concise representation h_{final} to the global server for attack classification. For reproducibility, Table II presents the TCN architecture hyperparameters. These parameters are selected to balance model capacity with the computational constraints of edge healthcare devices. To ensure statistical reliability, the experimental results are based on multiple independent runs using different random seeds. The selected sequence length balanced the temporal pattern capture and processing performance for the edge deployment across all three datasets.

C. Client-Side: Lightweight Temporal Features Extraction

The client-side processing is implemented using a lightweight TCN architecture that is optimized for resource-constrained edge devices, as illustrated in Figure 2. Sequential network traffic data is processed by the TCN in batches, as illustrated in Algorithm 1. Each batch consists of sequences (X_b, y_b) , with X_b representing input characteristics and y_b the matching labels [31]. The TCN employs dilated causal convolutions where output at time t depends only on inputs at times $t' \leq t$, with dilation factors increasing exponentially across layers. The TemporalBlock implements residual connections using dilated causal convolutional layers as shown in equation 5. It provides the complete architectural specification for reproducibility, with edge optimizations including multi-head design and parameter efficiency.

$$\begin{aligned} H_1 &= \text{ReLU}(\text{BN}(\text{Conv1D}_d(X))) \\ H_2 &= \text{ReLU}(\text{BN}(\text{Conv1D}_d(H_1))) \\ \text{Output} &= \text{ReLU}(H_2 + \text{Proj}(X)) \end{aligned} \quad (5)$$

D. Server-Side: Global Deep Neural Network

A Deep Neural Network (DNN) classifier is implemented on the server-side architecture to analyze the concise temporal features that obtain from client devices. The classification is performed by the server using the global model $f_s(\cdot; \Theta_s)$ after receiving the client representation $h_{\text{final}} \in \mathbb{R}^{B \times 256}$ [33]. Equation 10 is used to perform the transformation in the server-side classifier, which is constructed as a DNN with hidden dimensions.

$$\hat{y}_b \leftarrow f_s(h_{\text{final}}; \Theta_s^{(k)}) \quad (10)$$

The class probabilities for batch X_b are represented by $\hat{y}_b$. The network employs a sequential design that incorporates batch normalization and dropout after each hidden layer to enhance the stability of the training process and improve generalization. Equation 11 is employed to determine the forward propagation through the server network.

$$\begin{aligned} \text{Layer}_1 &= \text{BatchNorm}(\text{ReLU}(W_1 h_{\text{final}} + b_1)) \\ \text{Layer}_2 &= \text{BatchNorm}(\text{ReLU}(W_2 \text{Layer}_1 + b_2)) \\ \text{Layer}_3 &= \text{BatchNorm}(\text{ReLU}(W_3 \text{Layer}_2 + b_3)) \\ \hat{y}_b &= W_4 \text{Layer}_3 + b_4 \end{aligned} \quad (11)$$

where the learnable biases and weights of the server model Θ_s are denoted as b_i and W_i , respectively. We optimize it by applying the cross-entropy loss function with the L2 regularization equation 12. Equation 13 updates the server model parameters using gradient descent.

$$\mathcal{L} \leftarrow \text{CrossEntropy}(\hat{y}_b, y_b) + \lambda \|\Theta\|_2^2 \quad (12)$$

$$\Theta_s^{(k)} \leftarrow \Theta_s^{(k)} - \eta \nabla_{\Theta_s^{(k)}} \mathcal{L} \quad (13)$$

The server-side architecture ensures that only intermediate features are transferred, rather than sensitive healthcare data, preserving the privacy-protecting characteristic of the split learning framework. This method enables effective classification while maintaining patient confidentiality throughout the FL process.

E. Adaptive Split Federated Learning Framework

SFL divides the neural network among devices of the network and a central server, thus maintaining data privacy and avoiding computational load on the local devices. The federated learning hyperparameters are formally stated as $T = 50$ communication rounds, where each round follows $E = 1$ local training round, and a client participation rate $p = 1.0$ to ensure full client participation. They have been chosen strategically to provide a consistent and repeatable basis of the evaluation of the core contributions. In addition to the adaptive aggregation technique and split architecture based on TCNs, the hyperparameters are based on FL concepts designed to handle client drift. All local client updates are performed using the Adam optimizer with a learning rate of 0.001 and batch size 32. This arrangement generated consistent and high-quality results across all three cybersecurity datasets examined in our experiments.

Algorithm 2 XAI-EdgeSFL: Multi-Head TCN With Adaptive Split Federated Learning

Input: Healthcare dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^M$, clients N , rounds T , sequence length L , heads H

Output: Global parameters Θ_c^* , Θ_s^* with XAI explanations

Phase 1: System Initialization

Load $\mathcal{D}$, clean: $\mathcal{D}_{\text{clean}} \leftarrow \text{RemoveNaN}(\mathcal{D})$

Balance: $\mathcal{D}_{\text{balanced}} \leftarrow \text{SMOTE}(\mathcal{D}_{\text{clean}})$

Standardize: $\tilde{x}_i \leftarrow \frac{x_i - \mu}{\sigma}$

Split: $\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}} \leftarrow \text{TrainTestSplit}(\mathcal{D}_{\text{balanced}})$

Distribute: $\{\mathcal{D}_k\}_{k=1}^N \leftarrow \text{IIDPartition}(\mathcal{D}_{\text{train}}, N)$

Phase 2: Multi-Head TCN Architecture

Initialize client:

$f_c(\cdot; \Theta_c) \leftarrow \text{ClientSideTCN}(d_{\text{in}}, H, L)$

Initialize server:

$f_s(\cdot; \Theta_s) \leftarrow \text{ServerSideClassifier}(H \times d_{\text{hidden}})$

Initialize XAI: $\mathcal{X} \leftarrow \text{XAIAnalyzer}(f_c, f_s, \mathcal{F}, \mathcal{C})$

Phase 3: SFL Training with Multi-Head TCN

for $t = 1$ **to** T **do**

 Select clients: $\mathcal{S}_t \subset \{1, \dots, N\}$ with $|\mathcal{S}_t| = \lfloor pN \rfloor$

for each client $k \in \mathcal{S}_t$ **in parallel do**

$\Theta_c^{(t,k)}, \Theta_s^{(t,k)} \leftarrow$

$\text{ClientTCNUpdate}(k, \Theta_c^{(t-1)}, \Theta_s^{(t-1)})$

$\mathcal{A}_k \leftarrow \text{Evaluate}(\Theta_c^{(t,k)}, \Theta_s^{(t,k)}, \mathcal{D}_k)$

end

Multi-Head Adaptive Aggregation:

 Compute weights: $w_k^{(t)} = \frac{\exp(\mathcal{A}_k/\tau)}{\sum_{j \in \mathcal{S}_t} \exp(\mathcal{A}_j/\tau)}$

 Aggregate: $\Theta_c^{(t)} \leftarrow \sum_{k \in \mathcal{S}_t} w_k^{(t)} \Theta_c^{(t,k)}$

 Aggregate: $\Theta_s^{(t)} \leftarrow \sum_{k \in \mathcal{S}_t} w_k^{(t)} \Theta_s^{(t,k)}$

 Evaluate: $\mathcal{L}^{(t)}, \mathcal{A}^{(t)}, \mathcal{F}_1^{(t)} \leftarrow \text{Test}(\Theta_c^{(t)}, \Theta_s^{(t)}, \mathcal{D}_{\text{test}})$

end

Phase 4: Multi-Modal XAI Analysis

for $(x, y) \in \mathcal{D}_{\text{test}}^{\text{sample}}$ **do**

 Compute attributions:

$\phi(x) \leftarrow \mathcal{X}.\text{integrated_gradients_analysis}(x)$

 Generate explanations:

$\mathcal{E}(x) \leftarrow \{\phi_{IG}, \phi_{SHAP}, \phi_{LIME}\}$

end

return $\Theta_c^{(T)}, \Theta_s^{(T)}, \{\mathcal{E}(x)\}_{x \in \mathcal{D}_{\text{test}}^{\text{sample}}}$

1) *Federated Learning Optimization and Client Selection:* The framework coordinates distributed training among N clients over T federated rounds while protecting data privacy through selective client involvement, as described in Algorithm 2. Equation 14 determines which clients are active in each round t .

$$\mathcal{S}_t \subset \{1, \dots, N\} \quad \text{with} \quad |\mathcal{S}_t| = \lfloor pN \rfloor \quad (14)$$

Here, p represents the client participation rate. Each client $k \in \mathcal{S}_t$ begins with the global model parameters $(\Theta_c^{(t-1)}, \Theta_s^{(t-1)})$ and optimizes its personal dataset $\mathcal{D}_k$ using the client TCN

update process defined in equation 15.

$$\Theta_c^{(t,k)}, \Theta_s^{(t,k)} \leftarrow \text{ClientTCNUpdate}(k, \Theta_c^{(t-1)}, \Theta_s^{(t-1)}) \quad (15)$$

2) *Privacy-Preserving Split Architecture*: The framework implements a sophisticated split learning methodology that physically distributes the processing workload between the central server and client devices [5], [6]. A multi-head TCN model, as shown in equation 16, is employed to perform processing on the client side.

$$f_c(\cdot; \Theta_c) \leftarrow \text{ClientSideTCN}(d_{in}, H, L) \quad (16)$$

This design transforms the initial input data into useful intermediate patterns, while the classification process is carried out by the global server according to equation 17.

$$f_s(\cdot; \Theta_s) \leftarrow \text{ServerSideClassifier}(H \times d_{hidden}) \quad (17)$$

This architectural separation ensures sensitive healthcare data remains exclusively on client devices, with only privacy-preserving features being transmitted to the server [21].

3) *Performance-Based Aggregation Mechanism*: A dynamic aggregation mechanism is introduced, which is influenced by client performance metrics. Equation 18 is used to calculate aggregation weights when each client evaluates its model as $\mathcal{A}_k \leftarrow \text{Evaluate}(\Theta_c^{(t,k)}, \Theta_s^{(t,k)}, \mathcal{D}_k)$. $\mathcal{A}_k$ represents the local model accuracy of client k .

$$w_k^{(t)} = \frac{\exp(\mathcal{A}_k/\tau)}{\sum_{j \in \mathcal{S}_t} \exp(\mathcal{A}_j/\tau)} \quad (18)$$

where $\tau = 0.1$ denotes the temperature parameter that determines the sharpness of the softmax function. A smaller τ value emphasizes variations in client accuracies, providing higher-performing clients proportionally significant weights in the aggregate. This selective weighting helps the model converge faster while making it more resistant to clients that are either underperforming or acting maliciously. The global model then gets updated using a weighted average (Equation 19). We then run a full performance evaluation (Equation 20) to assess the results.

$$\Theta_c^{(t)} \leftarrow \sum_{k \in \mathcal{S}_t} w_k^{(t)} \Theta_c^{(t,k)}, \quad \Theta_s^{(t)} \leftarrow \sum_{k \in \mathcal{S}_t} w_k^{(t)} \Theta_s^{(t,k)} \quad (19)$$

$$\mathcal{L}^{(t)}, \mathcal{A}^{(t)}, \mathcal{F}_1^{(t)} \leftarrow \text{Test}(\Theta_c^{(t)}, \Theta_s^{(t)}, \mathcal{D}_{test}) \quad (20)$$

This method aggregates model updates in a way that improves robustness and speeds up convergence. It works by assigning higher importance to clients with strong performance, enabling the system to dynamically select the most dependable contributions each round. This steadily improves accuracy and stability. Together, split learning, adaptive client selection, and this performance-based approach make collaborative training secure and fast, even in highly distributed or remote scenarios.

IV. RESULTS AND DISCUSSIONS

A. Performance Metrics

We chose a variety of network traffic-based datasets and several clients to thoroughly test the proposed method. The suggested method is tested with three publicly available

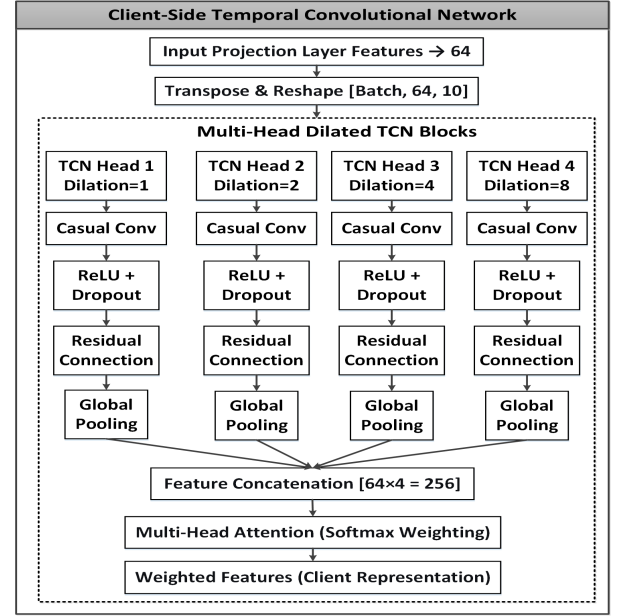


Fig. 2. Client-side TCN for feature extraction in SFL using dilated convolutions to capture temporal patterns in network traffic.

standard datasets: CICIoMT2024, N-BaIoT, and CICIDS2017. We evaluated performance by utilizing precision, recall, F-measure, and accuracy, respectively. These measures are derived from True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) values. Equation 21- 24 provides performance measurements.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (21)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (22)$$

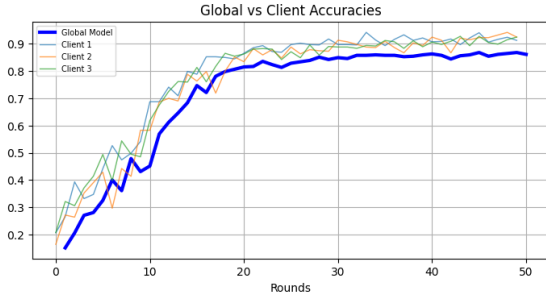
$$F\text{-measure} = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (23)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (24)$$

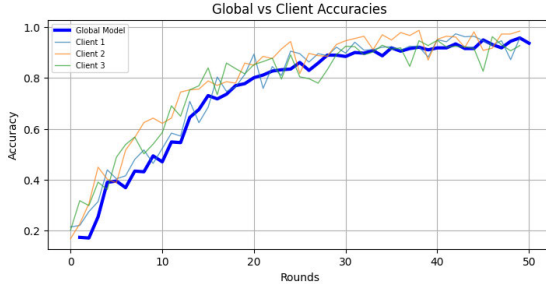
B. Performance Analysis and Comparisons

Figure 3 shows client accuracy rising from roughly 20% to over 92% for CICIoMT2024 and to 92–99% for N-BaIoT. The global models achieved 86.15% and 93.73%, respectively, demonstrating effective privacy-preserving learning. Figure 4 confirms substantial gains; client accuracy improved to 71–93% on CICIoMT2024 and to 85–97% on CICIDS2017, with global models reaching 79.15% and 92.37%. This highlights strong multi-client collaboration for security tasks. Figure 5 shows adaptive aggregation weights for three clients using CICIDS2017. Weights fluctuated between 0.24 and 0.41, with a maximum of 0.4020, indicating the framework rebalances the global model to prioritize higher-quality updates. Convergence is shown as weights stabilize, achieving global accuracies of 86.15%, 93.73%, and 99.46% while maintaining fairness. Figure 6 presents weights for 8 clients using CICIoMT2024.

Table III shows the proposed approach on CICIoMT2024, with three clients achieved strong performance. F1-scores

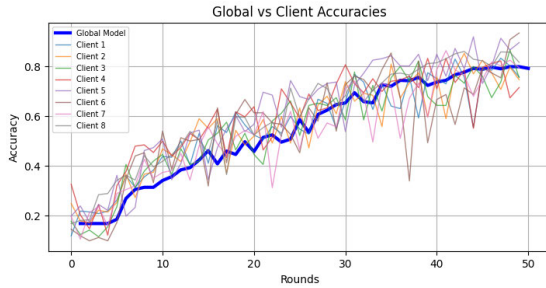


(a) CICIoMT2024

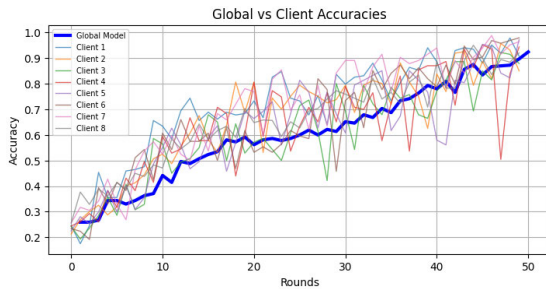


(b) N-BaIoT

Fig. 3. Accuracy curves of individual clients and the aggregated global model under the 3-client configuration using CICIoMT2024, N-BaIoT, and CICIDS2017.



(a) CICIoMT2024



(b) CICIDS2017

Fig. 4. Accuracy curves of individual clients and the aggregated global model under the 8-client configuration using CICIoMT2024, N-BaIoT, and CICIDS2017.

reached 0.97 for MQTT, 0.96 for Recon, and 0.92 for Spoofing attacks. While DDoS had a high recall at 0.98 with a lower precision of 0.60, detection was consistent overall. Similarly, Table IV for five clients shows maintained robustness. MQTT and Recon achieved F1-scores of 0.97 and 0.94, while the

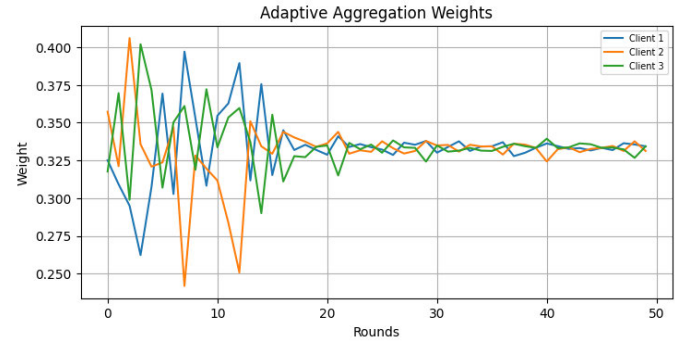


Fig. 5. Adaptive aggregated weights under the 3-client configuration using CICIDS2017 dataset.

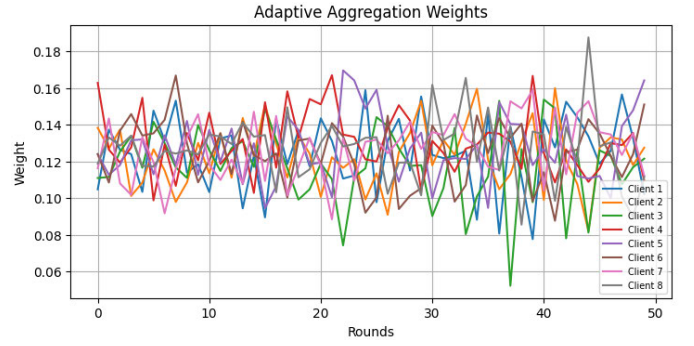


Fig. 6. Adaptive aggregated weights under the 8-client configuration using IoMT dataset.

TABLE III
PERFORMANCE ANALYSIS USING CICIoMT2024
DATASET WITH 3 CLIENTS

Class	Precision	Recall	F1-score
Benign	0.86	0.97	0.91
DDoS	0.60	0.98	0.74
DoS	0.98	0.49	0.65
MQTT	0.99	0.96	0.97
Recon	0.98	0.95	0.96
Spoofing	0.94	0.89	0.92

TABLE IV
PERFORMANCE ANALYSIS USING CICIoMT2024
DATASET WITH 5 CLIENTS

Class	Precision	Recall	F1-score
Benign	0.88	0.95	0.91
DDoS	0.60	0.90	0.72
DoS	0.85	0.51	0.64
MQTT	1.00	0.94	0.97
Recon	0.96	0.93	0.94
Spoofing	0.91	0.93	0.92

Benign and Spoofing classes scored 0.91 and 0.92, demonstrating reliable detection in a larger client setup.

Table V shows the model's performance on the N-BaIoT dataset with three clients. High classification accuracy was achieved for all attacks, with Scan and Syn classes obtaining F1-scores of 1.00 and 0.99, respectively, and UDP and UDP Plain classes scoring 0.96 and 0.89. The Ack class had a slightly lower F1-score of 0.82, yet the model remained robust across attack types. Table VI for five clients shows

TABLE V

PERFORMANCE ANALYSIS ON N-BAIoT DATASET WITH 3 CLIENTS

Class	Precision	Recall	F1-score
Ack	0.88	0.77	0.82
Scan	1.00	1.00	1.00
Syn	0.99	1.00	0.99
UDP	0.96	0.95	0.96
UDP Plain	0.85	0.94	0.89

TABLE VI

PERFORMANCE ANALYSIS ON N-BAIoT DATASET WITH 5 CLIENTS

Class	Precision	Recall	F1-score
Ack	0.76	0.77	0.76
Scan	0.99	0.99	0.99
Syn	0.99	1.00	0.99
UDP	0.93	0.86	0.89
UDP Plain	0.85	0.90	0.88

TABLE VII

PERFORMANCE ANALYSIS ON CICIDS2017 DATASET WITH 3 CLIENTS

Class	Precision	Recall	F1-score
Benign	0.99	0.99	0.99
Brute Force	1.00	1.00	1.00
SQL Injection	0.99	0.99	0.99
XSS	1.00	1.00	1.00

TABLE VIII

PERFORMANCE ANALYSIS ON CICIDS2017 DATASET WITH 5 CLIENTS

Class	Precision	Recall	F1-score
Benign	0.97	0.96	0.96
Brute Force	1.00	1.00	1.00
SQL Injection	0.96	0.97	0.97
XSS	1.00	1.00	1.00

TABLE IX

PERFORMANCE ANALYSIS ON CICIDS2017 DATASET WITH 8 CLIENTS

Class	Precision	Recall	F1-score
Benign	0.85	0.92	0.89
Brute Force	1.00	0.97	0.98
SQL Injection	0.94	0.84	0.89
XSS	0.90	0.98	0.94

consistent performance, with Scan and Syn classes achieving 0.99 F1-scores, UDP and UDP Plain at 0.89 and 0.88, and Ack at 0.76. Table VII shows that the suggested detection system demonstrates a very promising performance on the CICIDS2017 data set when the assessment is conducted on three client nodes. Specifically, the Brute-force and XSS attack groups scored the highest possible F1-score of 1.00, with the Benign traffic and the SQL-Injection category scoring an F1-score of 0.99. Table VIII shows the assessment of five clients using the given dataset. In line with the previous observations, the model has a high degree of detection accuracy for all classes. In particular, the F1-score of the benign and SQL-injection attacks was 0.96 and 0.97, respectively, whereas the brute-force and XSS attacks were again correctly identified with F1-scores of 1.00.

Table IX shows the performance analysis of the proposed model on the CICIDS2017 dataset with eight clients. The

findings show that, despite the increased client setup, the model retained high detection capacity across all classes. Brute Force and XSS achieved F1-scores of 0.98 and 0.94, while Benign and SQL Injection obtained 0.89. Table X evaluates the performance across various client settings using the CICIDS2017 dataset. A decline in global accuracy from 99.46% to 88.01% and an increase in global loss are observed as the number of clients increases from 3 to 10. As client heterogeneity rises in FL, model divergence in non-IID environments emerges. Heterogeneous data distributions across IoT devices cause local model updates to conflict, making global convergence challenging. XAI-EdgeSFL is resilient to decreases, maintaining competitive accuracy (88.01% with 10 clients) using adaptive aggregation that dynamically weights client contributions based on trust scores, minimizing divergence effects. The performance comparison between various existing FL-based research and the proposed XAI-EdgeSFL approach is shown in Table XI. XAI-EdgeSFL improves performance with a lightweight multi-head TCN for edge feature extraction, adaptive aggregation to prioritize reliable clients, and a multi-modal XAI suite for transparency and refinement.

The main performance trade-offs that are presented by the proposed architecture are shown in Table XII. The results show that privacy preservation incurs a major overhead of 6,360KB/round on communication and incurs inference time and client training time of 35.6 ms and 148 ms, respectively. This system has a steady round completion time of 0.63 seconds and an effective bandwidth utilization of 10,021 KB/Sec. This performance balance is appropriate in edge healthcare implementations, whereby privacy promotes better communication. The proposed XAI-EdgeSFL achieves 99.46% accuracy, supports local data processing to ensure privacy, integrates XAI tools to make the model explainable, and uses adaptive client weighting, demonstrated in Table R1P1. Centralized training has the best accuracy of 99.84%, although it requires the raw data to be aggregated, which may not be useful with distributed IoT, thus compromising data privacy. Compared to federated solutions, XAI-EdgeSFL is 10.31% better than the standard FL and 8.10% better than the traditional SFL. The unique combination of explainability and adaptability makes XAI-EdgeSFL the most compatible approach for secure and intelligible intrusion detection. The lightweight TCN is effective in edge deployments as shown by Table XIV. Compared to a typical TCN, the model achieves a higher accuracy (+ 0.34) and F1 score (+ 0.38) with only around 82% fewer parameters. It also provides approximately 75% faster inference with 35.6ms and less memory requirements of about 80% lower.

C. Multi-Modal Explainable AI Framework

We use a multi-modal XAI suite that includes SHAP, LIME, and IG because of their combined strengths in addressing the interpretability of our SFL architecture. IG provides reliable gradient-based attributions for deep TCN and DNN models, allowing for an effective understanding of feature contributions. SHAP generates consistent global feature importance values using cooperative game theory, allowing analysts to find

TABLE X
PERFORMANCE COMPARISON ACROSS DIFFERENT CLIENT CONFIGURATIONS USING CICIDS2017 DATASET

Clients	Global Accuracy	Global Loss	F1 (Weighted)	F1 (Macro)	Client Accuracies
3	0.9946	0.0345	0.9946	0.9947	1.0000, 1.0000, 0.9954
5	0.9809	0.0497	0.9809	0.9815	0.9590, 0.9957, 0.9893, 1.0000, 0.9933
8	0.9237	0.2306	0.9237	0.9240	0.9071, 0.8500, 0.8786, 0.9431, 0.9720, 0.9787, 0.9634, 0.9201
10	0.8801	0.3332	0.8800	0.8797	0.8014, 0.9545, 0.7627, 0.5421, 0.7188, 0.8605, 0.8974, 0.8429, 0.9048, 0.9208

TABLE XI
PERFORMANCE COMPARISON WITH RELATED PUBLISHED STUDIES

Work	Method	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Shen et al. [6]	Ring SFL	99.10	—	—	—
Latif et al. [35]	HMAC with SFL	98.06	99.53	100	99.76
Otoun et al. [23]	Transfer learning with SFL	98.50	—	—	—
CUI et al. [36]	SDN-based FL	99.12	—	—	99.20
Wenting et al. [15]	Personalized FL	98.1	92.1	88.1	88.9
Proposed Approach	XAI-EdgeSFL	99.46	99.00	100	99.47

Note: Ring SFL: Coordinates clients in a ring topology to reduce straggler effects. HMAC with SFL: Uses cryptographic authentication to secure client updates. Transfer learning with SFL: Fine-tunes a pre-trained model within a split federated architecture. SDN-based FL: Integrates federated learning with software-defined networking for dynamic aggregation. Personalized FL: Trains client-specific models while leveraging shared global knowledge. The proposed XAI-EdgeSFL outperforms existing approaches across key detection parameters with high accuracy, precision, recall, and a balanced F1-score.

TABLE XII
PERFORMANCE TRADE-OFF MEASUREMENTS FOR XAI-EDGE SFL

Metric	Mean	Std	Range	Unit
Communication	6360.13	0.00	6360.13-6360.13	KB
Duration	0.63	0.14	0.58-1.58	seconds
Training Time	0.15	0.01	0.14-0.17	seconds
Latency	35.62	10.01	19.98-59.98	milliseconds
Memory	673.84	16.90	555.61-677.50	MB
Bandwidth	10021.59	—	—	KB/sec

TABLE XIII
PERFORMANCE COMPARISON ON CIC-IDS2017 DATASET

Method	Accuracy (%)	F1-Score (%)	Δ vs XAI-EdgeSFL
Centrl: Train:	99.84	99.84	+0.38%
XAI-EdgeSFL	99.46	99.47	—
Traditional SFL	91.36	91.37	-8.10%
Standard FL	89.15	89.10	-10.31%

TABLE XIV
EFFICIENCY COMPARISON: LIGHTWEIGHT VS. STANDARD TCN

Metric	Lightweight TCN	Standard TCN	Improvement
Accuracy (%)	99.46	99.12	+0.34%
F1-score (%)	99.47	99.09	+0.38%
Parameters (M)	0.86	4.75	~82% reduction
Latency (ms)	35.6	142.8	~75% faster
Memory (MB)	673.8	3,369.0	~80% smaller

the most significant network indicators across the federated model. LIME provides locally interpretable, model-agnostic explanations for individual predictions. This feature is vital for analyzing specific instances of attacks, which helps build trust in medical professionals working with edge networks. Integrated Gradients and SHAP, which are designed for split-learning architectures, give more complex explanation methods than LIME, which delivers instance-level insights without internal model access. Cybersecurity specialists can utilize these tools to confirm the reasoning underlying threat detection and norm compliance criteria, as well as to design more intelligible algorithms that improve system confidence. These approaches are especially useful for identifying large

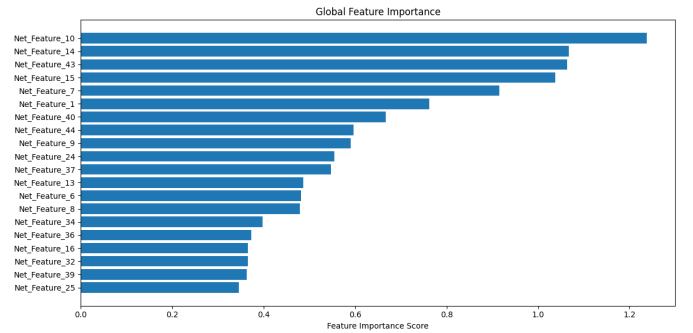


Fig. 7. Global features importance from CICIoMT2024 dataset.

feature contributions across many attack categories [36], [37]. The framework uses a multi-modal XAI suite, started as $\mathcal{X} \leftarrow \text{XAIAnalyzer}(f_c, f_s, \mathcal{F}, \mathcal{C})$. This incorporates the feature extractor f_c and the classifier f_s on the client side, as well as the feature labels $\mathcal{F}$ and the class labels $\mathcal{C}$. The system produces complementary explanations for each test sample $(x, y) \in \mathcal{D}_{\text{test}}^{\text{sample}}$ using several strategies as indicated in equations 25 and 26.

$$\phi(x) \leftarrow \mathcal{X}.\text{integrated_gradients_analysis}(x) \quad (25)$$

$$\mathcal{E}(x) \leftarrow \{\phi_{IG}, \phi_{SHAP}, \phi_{LIME}\} \quad (26)$$

The global feature importance for IDS is identified by the XAI analysis from our secure adaptive SFL framework, as illustrated in Figure 7. The model trained over 50 federation rounds and identified Net_Feature_10 (1.241), Net_Feature_15 (1.206), and Net_Feature_43 (1.168) as the most critical indications of malicious conduct. This feature hierarchy, which includes secondary contributors like Net_Feature_7 (1.087) and Net_Feature_14 (0.969), shows a distributed set of critical network characteristics for distinguishing between benign and malicious traffic. The analysis gives practical insights for protecting privacy and optimizing security monitoring and feature selection in FL.

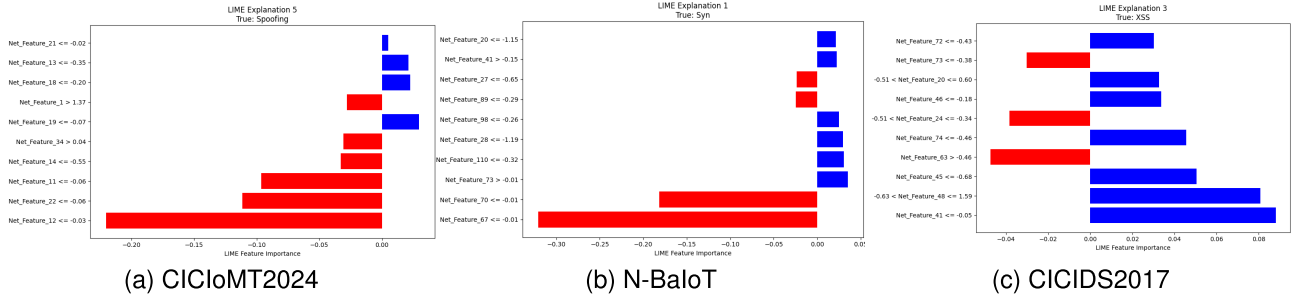


Fig. 8. LIME explanations for attack predictions, demonstrating feature contributions across attack classes. Each subplot shows the top attributes influencing the model's decision, with positive (blue) and negative (red) contributions indicating their relevance. Net_Feature_X labels identify network traffic aspects such as flow timing, packet size, and protocol characteristics based on their position in the preprocessed dataset.

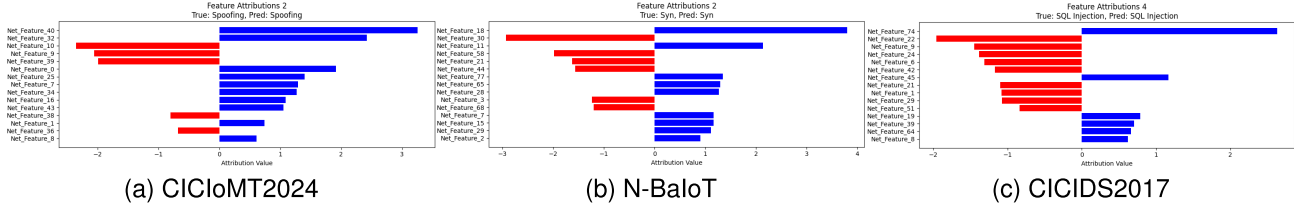


Fig. 9. IG-based feature attribution analysis showing network traffic feature relevance across attack classes with positive (blue) and negative (red) bars. Net_Feature_X feature indices from the processed dataset represent flow time, packet statistics, and protocol anomalies that influence model decisions.

Figure 8 shows LIME explanations for spoofing and syn attacks. For the spoofing attack, low values of features like Net_Feature_21, 13, 18, and 19 are shown as blue bars, increasing the prediction probability, while red bars decrease it. For the Syn attack, low values of Net_Feature_27, 89, 70, and 67 appear as red bars, lowering its probability, while features like Net_Feature_20 and 41 appear as blue bars, raising it. A similar analysis applies to XSS attacks on CICIDS2017. These color-coded visualizations clarify feature contributions, enabling analysts to verify model reliability, identify biases, and enhance detection accuracy.

Figure 9 shows the feature attributions of XAI of the spoofing and Syn attacks. In the case of the spoofing, there are red bars that depict features Net_Feature_10, 9, 39, 38, and 36, which influence the model prediction negatively. In contrast, the blue bars of Net_Feature_40, 32, and 0 show a contributing effect that is positive and increases confidence. The same trend can be seen with Syn attack, with Net_Feature_30, 58, 21, 44, 3, and 68 being red, which reduces the probability of the prediction, and Net_Feature_18, 11, and 77 being blue, which enhances it. The visualization can help an analyst confirm the reliability of the model, detect possible biases, and improve the detection capabilities in spoofing, Syn, and SQL-injection attacks. Server-side XAI analysis is conducted in the aggregation model globally during inference, thus eliminating computational overhead. Edge clients do not need to provide SHAP, LIME, or IG explanations during training or inference, which optimizes resource-limited edge devices to do real-time feature extraction and, at the same time, allows transparency and validation by the XAI suite.

V. CONCLUSION

The paper introduces a novel framework known as XAI-EdgeSFL of XAI and Adaptive Intrusion-Resilient SFL to promote security, privacy, and transparency in edge-based

healthcare. The methodology provides effective feature extraction and classification, and the sensitive data are stored locally by the implementation of lightweight TCN on consumer devices and more advanced classifiers on a global server. The adaptive aggregation scheme provides even deeper resilience to changing cyber threats, and the multimodal XAI framework, such as SHAP, LIME, and integrated gradients, provides understandable information enabling the building of clinical trust. The suggested method provides a strong protection against zero-day attacks based on a multimodal XAI package of SHAP, LIME, and Integrated Gradients. The suite may enable security professionals to query low-confidence predictions and identify abnormal traffic patterns, such as unknown types of attacks. The explainability-oriented, human-in-the-loop paradigm provides powerful investigation speed without the need to retrain the models. The resulting federated retraining cycle to implement a new class of attacks may be simplified in the future through lightweight continual learning methods. The cryptographic mechanisms can also be included in future research to protect the model updates and aggregation in the process of SF to contribute to privacy and trust. Potential study areas include homomorphic encryption, secure multi-party computation, and differential privacy. Such methods can be integrated to protect important updates from being intercepted or altered.

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